The humerus bone in the upper arm and the radius bone in the lower arm can be modeled as a lever with the elbow joint as the pivot point. Movement about the elbow results from a combination of internal and external forces acting on these bones. The primary internal force is generated from contractions of skeletal muscles; a 1-cm^2 cross-sectional area of skeletal muscle can generate a maximum of 7×10^6 dynes of force ($1 \text{ dyn} = 1 \text{ g}\cdot\text{cm/s}^2$). Friction between the humerus and the radius at the elbow is another internal force that can affect arm movements. As people age, cartilage degeneration decreases the lubrication between bones, which increases the coefficients of static and kinetic friction.

The contraction of the biceps, a skeletal muscle in the upper arm, exerts a pulling force on the radius in the lower arm. If the torque generated by this muscle exceeds that of any opposing internal and external forces, the arm curls about the elbow. This specific type of movement is known as a concentric muscle contraction.

The following experiments were performed with a 30-year-old male subject. Electrodes were placed around the subject's biceps and attached to an electromyograph, a device that measures the electrical activity of skeletal muscle contractions. The estimated weight of the subject's lower arm is 45 N, and the subject's lower arm segments to the elbow are shown in Table 1.

Table 1 Lower Arm Segment Lengths

Position	Distance from elbow (cm)
Biceps attachment	3
Lower arm center of gravity	20
Hand center of gravity	30
Tip of hand	45

Experiment 1

The subject kept his entire arm stationary and in a fixed position while holding balls of varying masses. The upper arm was perpendicular to the ground, and the lower arm was parallel to the ground (Figure 1).

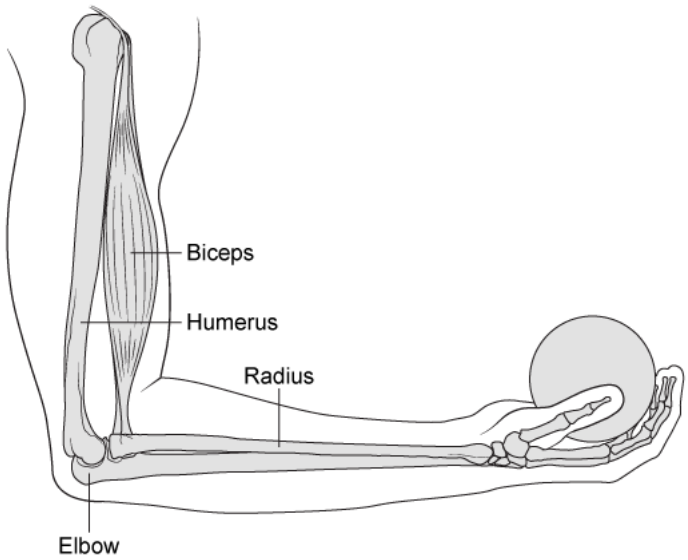


Figure 1 The fixed position of the arm during Experiment 1

Experiment 2

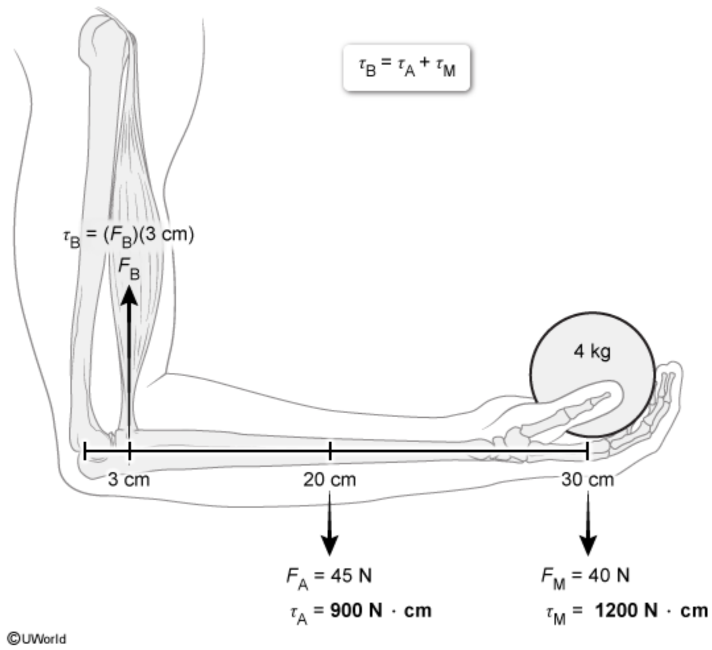
The subject kept his upper arm stationary and perpendicular to the ground while lifting and lowering balls of varying masses.

Suppose the subject is holding a 4-kg ball during Experiment 1, such that their arm is in static rotational equilibrium. Ignoring friction, what is the force exerted by the biceps? (Note: Use g = 10 m/s^2.)

- A. 85 N [37%]
- B. 340 N [25%]
- ✔ C. 700 N [31%]
- D. 2,100 N [6%]

Correct
31% answered correctly

Explanation:



In Experiment 1, the arm is held stationary while **three translational forces** (biceps force, weight of the arm, and weight of the ball) act on the lower arm, and each of these forces **each apply torque** (rotational force) about the elbow. The torque τ due to a force is the product of the lever arm length r (distance from pivot point) and the magnitude of the force F perpendicular to the lever arm:

$$\tau = rF$$

Because the arm is stationary, it is in **static rotational equilibrium**; there is no net rotational movement, and the sum of all torques is zero. Therefore, the torque due the biceps (B) is equal to the sum of the torques due to the weight of lower arm (A) and the ball's mass (M):

$$\tau_B = \tau_A + \tau_M$$

$$r_B F_B = r_A F_A + r_M F_M$$

The lever arm lengths are given in Table 1: r_B = 3 cm, r_A = 20 cm, and r_M = 30 cm. The force due to the lower arm is its weight given in the passage, F_A = 45 N. The weight of the ball is the product of its mass m

and the acceleration due to gravity *g*: $F_M = mg = (4\text{ kg})(10\text{ m/s}^2) = 40$ N. The **force exerted by the biceps** F_B is determined by rearranging the above equation and using these values:

$$(3\text{ cm})(F_B) = (20\text{ cm})(45\text{ N}) + (30\text{ cm})(40\text{ N}) = 2{,}100\text{ N}\cdot\text{cm}$$

$$F_B = \frac{2{,}100\text{ N}\cdot\cancel{\text{cm}}}{3\ \cancel{\text{cm}}} = 700\text{ N}$$

(Choice A) If the given translational forces are thought to be in static *translational equilibrium*, the incorrectly calculated force is 85 N. However, the translational normal force at the elbow joint is not given.

(Choice B) If the mass of the ball is used instead of its weight, the incorrectly calculated force is 340 N.

(Choice D) The *torque* exerted by the biceps is 2,100 N·cm. This value is divided by the lever arm length (3 cm) to obtain the force exerted by the biceps.

Educational objective:

Torque τ is a rotational force equal to the product of the lever arm length *r* and the perpendicular force *F*: $\tau = rF$. When in static rotational equilibrium, there is no net rotational movement, and the sum of all torques is zero.

Time spent:0 sec
Subject:
Physics

QID:400582
System:
Mechanics and Energy

The humerus bone in the upper arm and the radius bone in the lower arm can be modeled as a lever with the elbow joint as the pivot point. Movement about the elbow results from a combination of internal and external forces acting on these bones. The primary internal force is generated from contractions of skeletal muscles; a 1-cm^2 cross-sectional area of skeletal muscle can generate a maximum of 7×10^6 dynes of force (1 dyn = 1 g·cm/s^2). Friction between the humerus and the radius at the elbow is another internal force that can affect arm movements. As people age, cartilage degeneration decreases the lubrication between bones, which increases the coefficients of static and kinetic friction.

The contraction of the biceps, a skeletal muscle in the upper arm, exerts a pulling force on the radius in the lower arm. If the torque generated by this muscle exceeds that of any opposing internal and external forces, the arm curls about the elbow. This specific type of movement is known as a concentric muscle contraction.

The following experiments were performed with a 30-year-old male subject. Electrodes were placed around the subject's biceps and attached to an electromyograph, a device that measures the electrical activity of skeletal muscle contractions. The estimated weight of the subject's lower arm is 45 N, and the subject's lower arm segments to the elbow are shown in Table 1.

Table 1 Lower Arm Segment Lengths

Position	Distance from elbow (cm)
Biceps attachment	3
Lower arm center of gravity	20
Hand center of gravity	30
Tip of hand	45

Experiment 1

The subject kept his entire arm stationary and in a fixed position while holding balls of varying masses. The upper arm was perpendicular to the ground, and the lower arm was parallel to the ground (Figure 1).

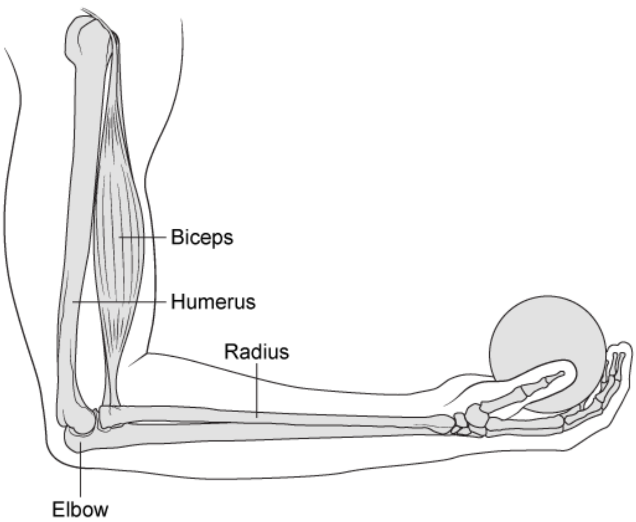


Figure 1 The fixed position of the arm during Experiment 1

Experiment 2

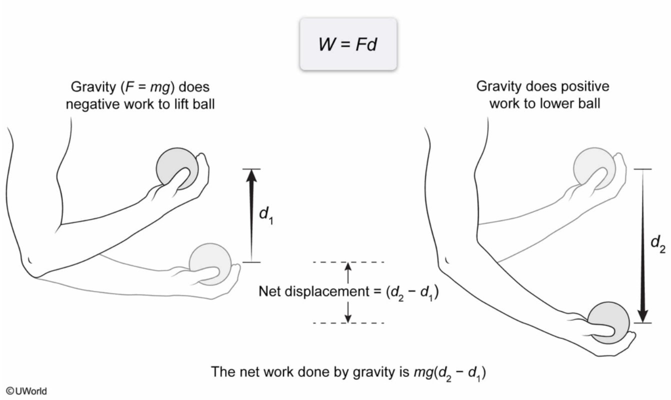
The subject kept his upper arm stationary and perpendicular to the ground while lifting and lowering balls of varying masses.

During Experiment 2, the subject lifts a ball with a mass m a vertical distance d_1 and then lowers the ball a greater vertical distance d_2. What is the net work done by gravity on the ball?

- A. $W = 0$ for all cases because gravity is a conservative force [15%]
- B. $W = mg(d_2 - d_1)$, because gravity does work to lift and lower the ball [45%]
- C. $W = mgd_2$, because gravity does work only to lower the ball [16%]
- D. $W = mg(d_1 + d_2)$, because gravity does work only on the net vertical path [23%]

Correct
44% answered correctly

Explanation:



Gravity is a **conservative force**; it depends on the initial and final position of the mass it acts on. The net **work** W done by a conservative force is the

amount of energy transferred by the force F over a displacement d in the direction of the force:

$$W = Fd$$

Work and displacement are positive if they are in the same direction as the force, and gravity does work only along the vertical axis; W is positive when the ball is lowered because gravity acts downward. The ball's **net downward displacement** is the difference between the lowered distance d_2 and the raised distance d_1:

$$d = d_2 - d_1$$

$$W = F(d_2 - d_1)$$

The **force due to gravity** on the ball is its weight, which is the product of its mass m and the acceleration due to gravity g: $F = mg$. Therefore, the net work done by gravity is

$$W = mg(d_2 - d_1)$$

(Choice A) Although the statement "gravity is a conservative force" is accurate, the work done by gravity is not always zero. $W = 0$ if and only if there is no net displacement in the vertical direction.

(Choice C) The expression mgd_2 is the positive work gravity does to lower the ball. However, gravity also does negative work ($-mgd_1$) when the ball is lifted. Work is negative if the force opposes the object's motion.

(Choice D) Although the statement "gravity does work only on the net vertical path" is accurate, the expression $mg(d_1 + d_2)$ is not the net displacement of the ball but the *total distance* traveled.

Educational objective:
Gravity is a conservative force, and the work W done by a conservative force F over a displacement d is $W = Fd$, where d is in the same direction as the force. The force due to gravity (weight) is the product of mass m and gravitational acceleration g.

Time spent:0 sec
Subject:
Physics

QID:400583
System:
Mechanics and Energy

The humerus bone in the upper arm and the radius bone in the lower arm can be modeled as a lever with the elbow joint as the pivot point. Movement about the elbow results from a combination of internal and external forces acting on these bones. The primary internal force is generated from contractions of skeletal muscles; a 1-cm^2 cross-sectional area of skeletal muscle can generate a maximum of 7×10^6 dynes of force (1 dyn = 1 $g \cdot cm/s^2$). Friction between the humerus and the radius at the elbow is another internal force that can affect arm movements. As people age, cartilage degeneration decreases the lubrication between bones, which increases the coefficients of static and kinetic friction.

The contraction of the biceps, a skeletal muscle in the upper arm, exerts a pulling force on the radius in the lower arm. If the torque generated by this muscle exceeds that of any opposing internal and external forces, the arm curls about the elbow. This specific type of movement is known as a concentric muscle contraction.

The following experiments were performed with a 30-year-old male subject. Electrodes were placed around the subject's biceps and attached to an electromyograph, a device that measures the electrical activity of skeletal muscle contractions. The estimated weight of the subject's lower arm is 45 N, and the subject's lower arm segments to the elbow are shown in Table 1.

Table 1 Lower Arm Segment Lengths

Position	Distance from elbow (cm)
Biceps attachment	3
Lower arm center of gravity	20
Hand center of gravity	30
Tip of hand	45

Experiment 1

The subject kept his entire arm stationary and in a fixed position while holding balls of varying masses. The upper arm was perpendicular to the ground, and the lower arm was parallel to the ground (Figure 1).

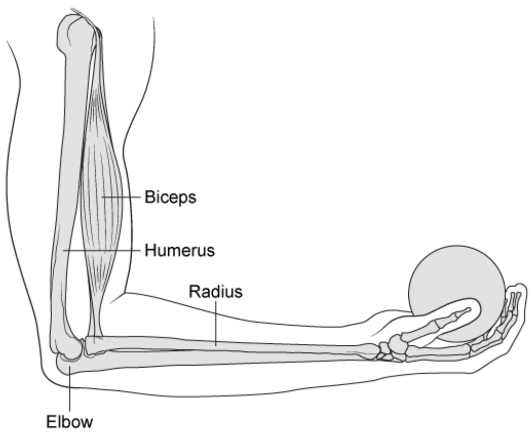


Figure 1 The fixed position of the arm during Experiment 1

Experiment 2

The subject kept his upper arm stationary and perpendicular to the ground while lifting and lowering balls of varying masses.

What is the maximum pressure in N/m^2 generated by skeletal muscles?

- A. 7×10^3 N/m^2 [26%]
- B. 7×10^5 N/m^2 [46%]
- C. 7×10^6 N/m^2 [16%]
- D. 7×10^9 N/m^2 [10%]

Correct
46% answered correctly

Explanation:

Convert units from dyn/cm^2 to N/m^2 through base SI units

Derived units	$7 \times 10^6 \frac{\text{dyn}}{\text{cm}^2}$ ↓	$1 \frac{\text{N}}{\text{m}^2}$ ↓
SI units	$7 \times 10^6 \frac{\text{g}}{\text{cm} \cdot \text{s}^2}$ ↓	
Base SI units	$7 \times 10^5 \frac{\textbf{kg}}{\textbf{m} \cdot \textbf{s}^2}$	$1 \frac{\textbf{kg}}{\textbf{m} \cdot \textbf{s}^2}$

Pressure *P* is defined as the ratio of the applied force *F* and the area *A* that is perpendicular to *F*:

$$P = \frac{F}{A}$$

According to the passage, a 1-cm^2 cross-sectional area of skeletal muscle can generate 7×10^6 dynes (dyn) of force. Therefore, the maximum *P* is:

$$P = \frac{7 \times 10^6 \text{ dyn}}{1 \text{ cm}^2} = 7 \times 10^6 \frac{\text{dyn}}{\text{cm}^2}$$

The units $\frac{\text{dyn}}{\text{cm}^2}$ and $\frac{\text{N}}{\text{m}^2}$ both represent *P* using derived units of force (dynes, newtons). The above value can be converted to its equivalent value in $\frac{\text{N}}{\text{m}^2}$ through **dimensional analysis**, which uses the relationship between derived units, SI units (eg, gram, centimeter), and base SI units (eg, kilogram, meter).

The SI unit of *F*, the newton (N), is equal to $1\frac{\text{kg} \cdot \text{m}}{\text{s}^2}$. Using this conversion, the base SI units for *P*, $\frac{\text{N}}{\text{m}^2}$, are:

$$\frac{\text{N}}{\text{m}^2} = \left(\frac{\cancel{\text{N}}}{\text{m}^2}\right)\left(\frac{\frac{\text{kg}\cdot\text{m}}{\text{s}^2}}{\cancel{\text{N}}}\right) = \frac{\text{kg}}{\text{m}\cdot\text{s}^2}$$

Therefore, any value of *P* converted to the above base SI units will be its equivalent value in $\frac{\text{N}}{\text{m}^2}$. Before $7 \times 10^6 \frac{\text{dyn}}{\text{cm}^2}$ can be converted to base SI units, it is rewritten **in SI units** using the conversion provided in the passage, $1\text{ dyn} = 1\frac{\text{g}\cdot\text{cm}}{\text{s}^2}$:

$$P = \left(7 \times 10^6 \frac{\cancel{\text{dyn}}}{\text{cm}^2}\right)\left(\frac{\frac{\text{g}\cdot\text{cm}}{\text{s}^2}}{\cancel{\text{dyn}}}\right) = 7 \times 10^6 \frac{\text{g}\cdot\text{cm}}{\text{cm}^2\cdot\text{s}^2} = 7 \times 10^6 \frac{\text{g}}{\text{cm}\cdot\text{s}^2}$$

This value is rewritten **in base SI units** (1,000 g = 1 kg and 100 cm = 1 m):

$$P = \left(7 \times 10^6 \frac{\cancel{\text{g}}}{\text{cm}\cdot\text{s}^2}\right)\left(\frac{1\text{ kg}}{1{,}000\text{ }\cancel{\text{g}}}\right) = 7 \times 10^3 \frac{\text{kg}}{\text{cm}\cdot\text{s}^2}$$

$$P = \left(7 \times 10^3 \frac{\text{kg}}{\cancel{\text{cm}}\cdot\text{s}^2}\right)\left(\frac{100\text{ }\cancel{\text{cm}}}{1\text{ m}}\right) = 7 \times 10^5 \frac{\text{kg}}{\text{m}\cdot\text{s}^2}$$

As determined above, the unit $\frac{\text{kg}}{\text{m}\cdot\text{s}^2}$ is equivalent to $\frac{\text{N}}{\text{m}^2}$, and therefore the maximum *P* generated by skeletal muscles has the same magnitude as the above value, $7 \times 10^5 \frac{\text{N}}{\text{m}^2}$.

(Choice A) If the conversion of cm^2 to m^2 is incorrectly performed as 100 cm^2 = 1 m^2, the result is 7×10^3 N/m^2. The correct conversion is 10,000 cm^2 = 1 m^2.

(Choice C) If the SI units (g, cm) are not converted to base SI units (kg, m), the result is 7×10^6 N/m^2.

(Choice D) If the pressure calculation step (dividing force by 1 cm^2) is skipped, the result is 7×10^9 N/m^2.

Educational objective:
Dimensional analysis is used to convert values from one set of units to another using the relationship between derived units, SI units, and base SI units. Pressure is calculated as force divided by area.

Time spent:0 sec
Subject:
Physics

QID:400584
System:
Mechanics and Energy

The humerus bone in the upper arm and the radius bone in the lower arm can be modeled as a lever with the elbow joint as the pivot point. Movement about the elbow results from a combination of internal and external forces acting on these bones. The primary internal force is generated from contractions of skeletal muscles; a 1-cm^2 cross-sectional area of skeletal muscle can generate a maximum of 7×10^6 dynes of force (1 dyn = 1 g·cm/s^2). Friction between the humerus and the radius at the elbow is another internal force that can affect arm movements. As people age, cartilage degeneration decreases the lubrication between bones, which increases the coefficients of static and kinetic friction.

The contraction of the biceps, a skeletal muscle in the upper arm, exerts a pulling force on the radius in the lower arm. If the torque generated by this muscle exceeds that of any opposing internal and external forces, the arm curls about the elbow. This specific type of movement is known as a concentric muscle contraction.

The following experiments were performed with a 30-year-old male subject. Electrodes were placed around the subject's biceps and attached to an electromyograph, a device that measures the electrical activity of skeletal muscle contractions. The estimated weight of the subject's lower arm is 45 N, and the subject's lower arm segments to the elbow are shown in Table 1.

Table 1 Lower Arm Segment Lengths

Position	Distance from elbow (cm)
Biceps attachment	3
Lower arm center of gravity	20
Hand center of gravity	30
Tip of hand	45

Experiment 1

The subject kept his entire arm stationary and in a fixed position while holding balls of varying masses. The upper arm was perpendicular to the ground, and the lower arm was parallel to the ground (Figure 1).

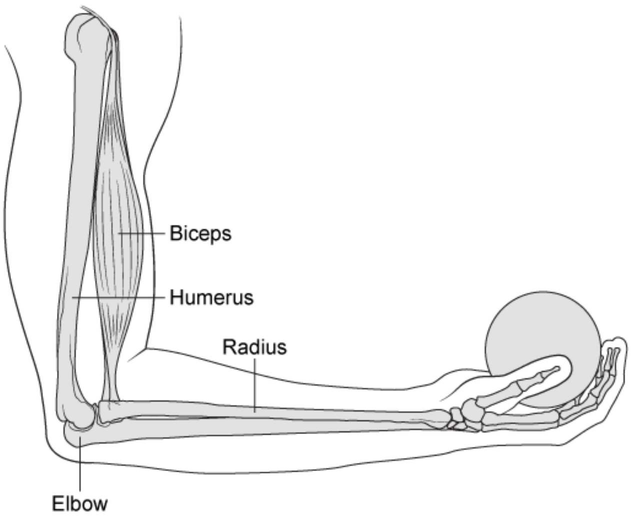


Figure 1 The fixed position of the arm during Experiment 1

Experiment 2

The subject kept his upper arm stationary and perpendicular to the ground while lifting and lowering balls of varying masses.

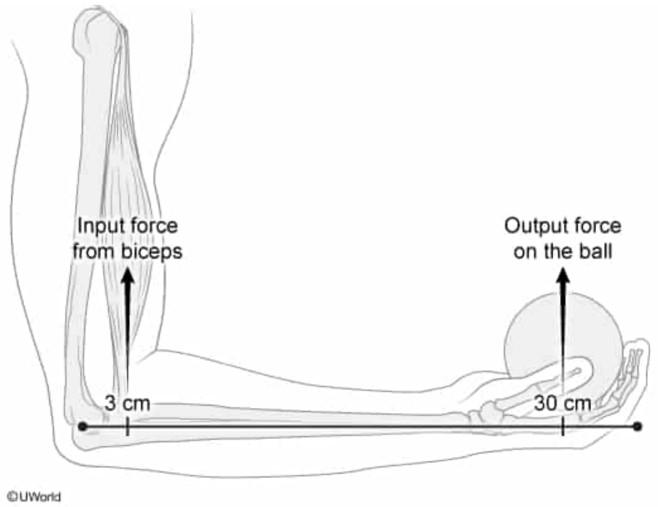
Ignoring the weight of the arm, what is the mechanical advantage of the biceps when it is lifting a ball in Experiment 2?

- A. 0.07 [8%]
- ✔ B. 0.10 [33%]
- C. 10 [46%]
- D. 15 [11%]

Correct
33% answered correctly

Explanation:

$$\text{Mechanical advantage} = \frac{F_o}{F_i} = \frac{d_i}{d_o}$$



According to the passage, the arm is modeled as a lever with the pivot point (fulcrum) at the elbow. **Mechanical advantage** is the ratio of the output force F_o to the input force F_i:

$$\text{mechanical advantage} = \frac{F_\text{o}}{F_\text{i}}$$

Work W is the amount of energy transferred by a force F over a distance d: $W = Fd$. A simple machine can amplify an input force through the **conservation of energy**; eg, a lever can transfer small forces over long distances into large forces over short distances. The work done by the output force must equal the work done by the input force:

$$F_\text{i}d_\text{i} = F_\text{o}d_\text{o}$$

where d_i and d_o are the distances from the pivot point to the input and output forces, respectively. These distances can be used in place of the vertical displacements of the forces due to the geometry of a lever. Rearranging the above equation, the mechanical advantage of a lever is equal to the **ratio of the input distance to the output distance**:

$$\text{mechanical advantage} = \frac{F_\text{o}}{F_\text{i}} = \frac{d_\text{i}}{d_\text{o}}$$

The question asks for the mechanical advantage of the subject's biceps in lifting a ball. The input distance d_i is 3 cm to the biceps, and the output distance d_o is 30 cm to the ball. Using these values, the mechanical advantage is

$$\frac{d_\text{i}}{d_\text{o}} = \frac{3\text{ cm}}{30\text{ cm}} = 0.1$$

The mechanical advantage of the biceps is <1, which means that a relatively large input force from the biceps is required to balance an opposing force at the hand.

(Choices A and D) If the entire length of the arm (45 cm) is incorrectly used as the output distance, the mechanical advantage of the biceps would be calculated as 0.07, and the inverse of this value is 15.

(Choice C) The mechanical advantage of *ball on the biceps* is 10 and is calculated by designating the ball as the input force and the biceps as the output force.

Educational objective:
The mechanical advantage of a simple machine is the ratio of the output force to the input force. Due to the conservation of energy, the mechanical advantage of a lever is also equal to the ratio of the input distance to the output distance.

Time spent: 0 sec
Subject: Physics

QID: 400585
System: Mechanics and Energy

The humerus bone in the upper arm and the radius bone in the lower arm can be modeled as a lever with the elbow joint as the pivot point. Movement about the elbow results from a combination of internal and external forces acting on these bones. The primary internal force is generated from contractions of skeletal muscles; a 1-cm^2 cross-sectional area of skeletal muscle can generate a maximum of 7×10^6 dynes of force (1 dyn = 1 g·cm/s^2). Friction between the humerus and the radius at the elbow is another internal force that can affect arm movements. As people age, cartilage degeneration decreases the lubrication between bones, which increases the coefficients of static and kinetic friction.

The contraction of the biceps, a skeletal muscle in the upper arm, exerts a pulling force on the radius in the lower arm. If the torque generated by this muscle exceeds that of any opposing internal and external forces, the arm curls about the elbow. This specific type of movement is known as a concentric muscle contraction.

The following experiments were performed with a 30-year-old male subject. Electrodes were placed around the subject's biceps and attached to an electromyograph, a device that measures the electrical activity of skeletal muscle contractions. The estimated weight of the subject's lower arm is 45 N, and the subject's lower arm segments to the elbow are shown in Table 1.

Table 1 Lower Arm Segment Lengths

Position	Distance from elbow (cm)
Biceps attachment	3
Lower arm center of gravity	20
Hand center of gravity	30
Tip of hand	45

Experiment 1

The subject kept his entire arm stationary and in a fixed position while holding balls of varying masses. The upper arm was perpendicular to the ground, and the lower arm was parallel to the ground (Figure 1).

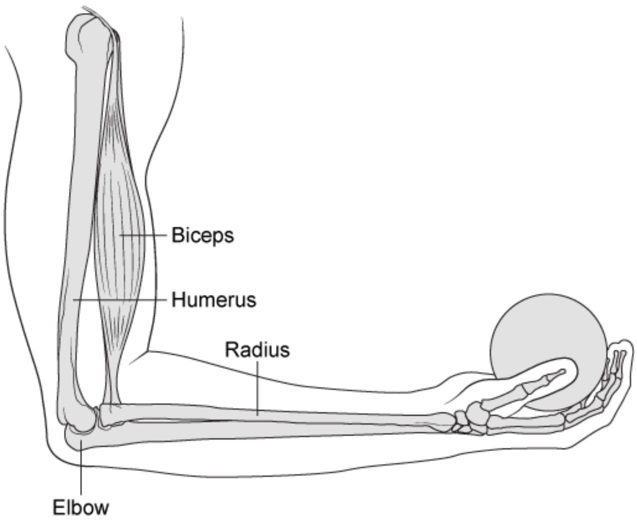


Figure 1 The fixed position of the arm during Experiment 1

Experiment 2

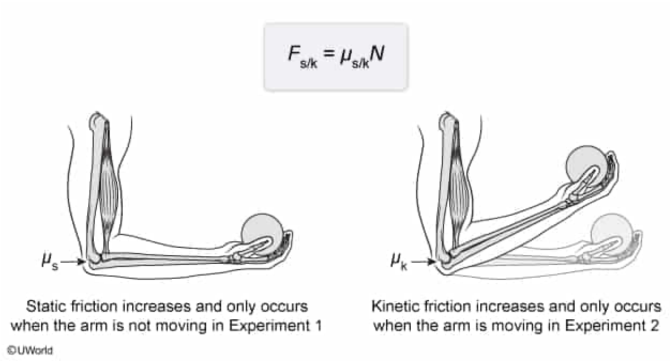
The subject kept his upper arm stationary and perpendicular to the ground while lifting and lowering balls of varying masses.

Assuming the weight of the arm remains the same, how would the friction at the elbow change if the subject were to repeat the experiments 20 years later?

- A. The maximum static friction during Experiment 1 would increase. [66%]
- B. The kinetic friction during Experiment 1 would increase. [19%]
- C. The maximum static friction during the lifting motion in Experiment 2 would decrease. [6%]
- D. The kinetic friction during the lifting motion in Experiment 2 would decrease. [7%]

Correct
67% answered correctly

Explanation:



Friction is the force that resists sliding between two surfaces, and it can be either static or kinetic. **Static friction** is involved when no sliding occurs, and **kinetic friction** is involved when sliding occurs. The maximum value of the static friction F_s (ie, just before sliding occurs) is defined as the product of the coefficient of static friction μ_s and the normal force N

$$F_s = \mu_s N$$

Similarly, the kinetic friction F_k is always the product of the coefficient of kinetic friction μ_k and N:

$$F_k = \mu_k N$$

The passage states that cartilage degeneration increases with age, resulting in an **increase** in both the static and kinetic **friction coefficients**. Therefore, the older subject will experience increased static and kinetic friction at the elbow. Because the arm is stationary during **Experiment 1**, the maximum **static frictional force would increase**.

(Choice B) Although kinetic friction is expected to increase due to cartilage degeneration, there is no movement in Experiment 1; therefore, there will be no kinetic friction.

(Choices C and D) Both static friction and kinetic friction are expected to increase because the coefficients of each will increase. In addition, there is no static friction at the elbow when the arm is moving.

Educational objective:
Static and kinetic friction are forces that resist sliding between two surfaces. Static friction occurs only when the two surfaces do not slide, and kinetic friction occurs only when the two surfaces slide relative to each other. The

magnitude of each depends on their respective coefficients of friction; therefore, an increase in the coefficients will increase the frictional force.

Time spent: 0 sec	**QID**:400586
Subject:	**System:**
Physics	Mechanics and Energy

The humerus bone in the upper arm and the radius bone in the lower arm can be modeled as a lever with the elbow joint as the pivot point. Movement about the elbow results from a combination of internal and external forces acting on these bones. The primary internal force is generated from contractions of skeletal muscles; a 1-cm^2 cross-sectional area of skeletal muscle can generate a maximum of 7×10^6 dynes of force (1 dyn = 1 g·cm/s^2). Friction between the humerus and the radius at the elbow is another internal force that can affect arm movements. As people age, cartilage degeneration decreases the lubrication between bones, which increases the coefficients of static and kinetic friction.

The contraction of the biceps, a skeletal muscle in the upper arm, exerts a pulling force on the radius in the lower arm. If the torque generated by this muscle exceeds that of any opposing internal and external forces, the arm curls about the elbow. This specific type of movement is known as a concentric muscle contraction.

The following experiments were performed with a 30-year-old male subject. Electrodes were placed around the subject's biceps and attached to an electromyograph, a device that measures the electrical activity of skeletal muscle contractions. The estimated weight of the subject's lower arm is 45 N, and the subject's lower arm segments to the elbow are shown in Table 1.

Table 1 Lower Arm Segment Lengths

Position	Distance from elbow (cm)
Biceps attachment	3
Lower arm center of gravity	20
Hand center of gravity	30
Tip of hand	45

Experiment 1

The subject kept his entire arm stationary and in a fixed position while holding balls of varying masses. The upper arm was perpendicular to the ground, and the lower arm was parallel to the ground (Figure 1).

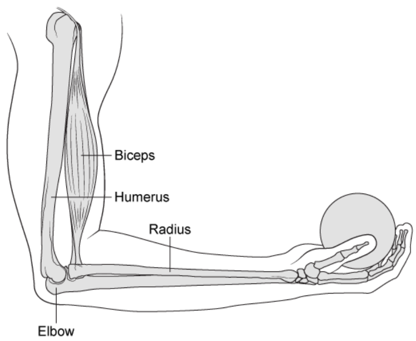


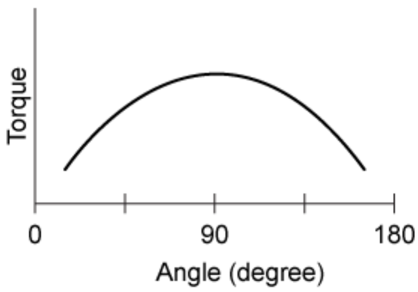
Figure 1 The fixed position of the arm during Experiment 1

Experiment 2

The subject kept his upper arm stationary and perpendicular to the ground while lifting and lowering balls of varying masses.

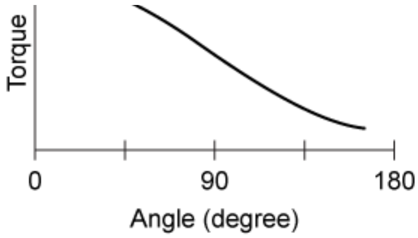
Which of the following graphs shows the relationship between the torque generated by the weight of the ball and the angle the biceps makes with the lower arm in Experiment 2?

A.



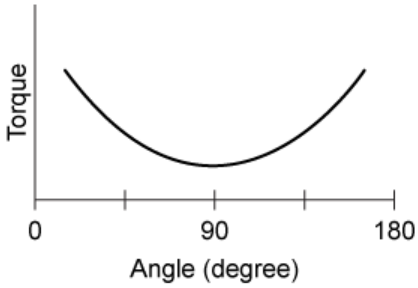
[70%]

B.



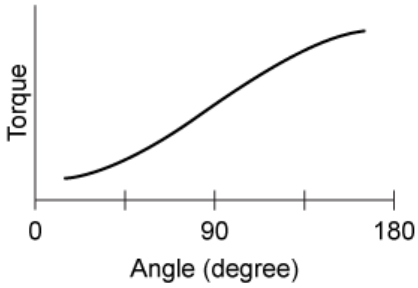
[7%]

○ C.



[11%]

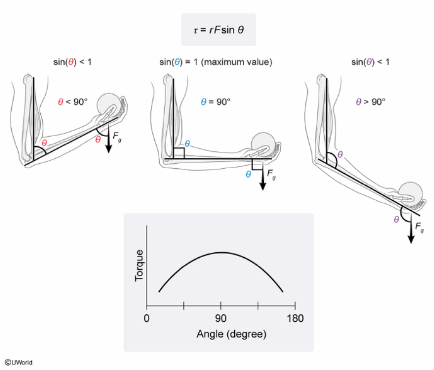
○ D.



[9%]

Correct
71% answered correctly

Explanation:



In Experiment 2, the subject lifts a ball while the biceps is perpendicular to the ground. The weight of the ball is always perpendicular to the ground and generates a **torque** (rotational force) about the elbow. The magnitude of the torque τ about the elbow is the product of the distance r to the elbow and the perpendicular component of the ball's weight F:

$$\tau = rF\sin\theta_{\text{ball}}$$

where θ_{ball} is the angle between the ball's weight and the lower arm. However, the question asks about the angle θ_{biceps} between the biceps and the lower arm. Because both the biceps and the weight of the ball are perpendicular to the ground, they are parallel to each other and these two angles are equivalent. In other words, the angle between the ball's weight and the arm is 30° if the angle at the biceps is also 30°.

The **sine function** takes the **perpendicular component** of a vector, and its magnitude is greatest when θ is 90°. In this case, sin θ determines the degree to which the ball's weight is perpendicular to the lower arm. When the lower arm is raised ($\theta < 90°$) or lowered ($\theta > 90°$), the ball's weight becomes "less perpendicular" to the lower arm, decreasing the magnitude of the torque on the elbow **(Choices B and D)**. Therefore, the correct graph will have a **maximum value at 90°** when the ball's weight is perpendicular to the lower arm.

(Choice C) The cosine function takes the *parallel* component of a vector, and its magnitude is lowest at 90°. However, torque is due to the perpendicular component of the force, and therefore depends on the sine function.

Educational objective:

Torque τ is the product of the distance r from the pivot point and the perpendicular component of the force F: $\tau = rF\sin\theta$, where θ is the angle between the force and the lever arm. The magnitude of $\sin\theta$ (and torque) is greatest when θ is 90°.

Time spent: 0 sec	**QID**:400587
Subject: Physics	**System:** Mechanics and Energy